

No fixed points on two points: a disproof of Conjecture

00000000459

gaochengzhecpu (AI-assisted submission)

3 October 2026

The claim and the counterexample. The conjecture asserts a rank-preserving bijection between the fixed points of Kreweras complementation on NC_n and certain 3-noncrossing partitions without isolated points. The asserted bijection already fails for $n = 2$: the first set is empty and the second contains exactly one object.

Definitions. A set partition is represented by the equivalence relation of belonging to the same block. A partition is noncrossing if there are no $a < b < c < d$ with a, c in one block and b, d in a different block. For two noncrossing partitions P, Q on $[n]$, interleave their vertices as

$$1, 1', 2, 2', \dots, n, n',$$

putting P on the unprimed positions and Q on the primed positions. The standard Kreweras complement $K(P)$ is the coarsest Q for which this combined partition is noncrossing; see Definitions 2.9, 2.17, and 2.19 of [1].

For the other side, an isolated point is a singleton block. In the standard arc representation, an arc joins consecutive elements of the same block. A 3-crossing requires three arcs $(a, d), (b, e), (c, f)$ with $a < b < c < d < e < f$; a 3-noncrossing partition has no such configuration. On two vertices this condition is automatic.

Compute the genuine complement. There are exactly two partitions of $\{1, 2\}$:

$$D = \{\{1\}, \{2\}\}, \quad T = \{\{1, 2\}\}.$$

Both are noncrossing. If $P = D$, the primed block $\{1', 2'\}$ creates no crossing, so its coarsest admissible complement is T . If $P = T$, choosing $Q = T$ would give alternating distinct blocks $\{1, 2\}$ and $\{1', 2'\}$ in the order $1, 1', 2, 2'$. Thus $Q = T$ is forbidden, while $Q = D$ is allowed. Consequently

$$K(D) = T, \quad K(T) = D.$$

Neither partition is fixed. This also shows directly that the map is an involution in this instance; no general involution assertion is needed.

On the target side, the partition T has no singleton blocks and is 3-noncrossing. It is the only such partition, since the other partition D has singleton blocks. If the phrase is read as noncrossing partitions without singletons, or as noncrossing matchings without isolated vertices, the same two-vertex object $\{1, 2\}$ still witnesses nonemptiness. Thus there cannot even be a surjection from the empty fixed-point set to the target set. A bijection, with or without a rank condition, is impossible. $\square$

Scope. We use the standard Kreweras complement and the ordinary meaning of partitions on $[n]$. The argument uses actual objects at $n = 2$, not a guessed formula for the Riordan numbers. The source imposes no exception for this value of n . The accompanying files are prepared for independent review.

Formal verification and exhaustive coverage. The accompanying Lean 4.19.0 project imports only `Std`. `Partition n` stores an arbitrary proposition-valued binary relation on `Fin n` with proofs of reflexivity, symmetry, and transitivity. It is therefore an actual set partition, rather than a label assigned to a preselected example.

`every_partition_encoded` proves that every partition of two points is exactly one of `model false` and `model true`, which are D and T . `model_injective` proves they are distinct, and `every_partition_noncrossing` verifies their eligibility for NC_2 . Extensional equality is equality of their full membership relations.

`interleave` implements the four positions $1, 1', 2, 2'$ using even and odd indices, with integer division by two giving the original vertex index. `interleave_is_partition` proves directly that the combined relation is an equivalence relation for any two input partitions. `Noncrossing` is the alternating-four-vertex condition above.

`IsKrewerasComplement P Q` requires both input partitions and their interleaving to be noncrossing, and requires every admissible partition to refine Q . Its maximality quantifier ranges over *all* proposition-valued partitions, using the proved complete classification when finite checking is appropriate. `kreweras_models` verifies the two complements by this definition. Existence for every input and uniqueness of the complement are proved separately. Hence `no_fixed_points` concerns the fixed points of the actual uniquely defined operation, not a substitute Boolean map.

The target uses the consecutive-block-element definition of an arc, the six-vertex 3-crossing condition, and the absence of singleton blocks. `targetWitness` constructs T as a valid target object, and `target_unique` proves that every target object is this one. The theorem `conjecture459_refuted` denies the existence of a surjective function from the actual fixed-point subtype to that target subtype. This is a necessary condition for the asserted rank-preserving bijection, so no formalization of an unused rank function is needed.

All finite decisions are kernel checked. There are no admitted proofs, additional axioms, or native decision procedures. The final theorem uses only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The Lake configuration treats warnings as errors.

References

- [1] K. Ebrahimi-Fard and T. Ringeard, *Planar Binary Trees, Noncrossing Partitions and the Operator-Valued S -Transform*, *Annals of Combinatorics* **29** (2025), 799–836. Definitions 2.9, 2.17, 2.19. <https://doi.org/10.1007/s00026-024-00730-1>.